QML Lab • research project 09

# Quantum Transfer Learning

A pretrained ResNet as feature extractor and a small quantum circuit as the classifier head.

Living document • last updated October 25, 2026

| | | | |
|---|---|---|---|
| **Quadrant** | CQ (classical data, quantum processor) | **Track** | Hybrid |
| **Difficulty** | Easy | **Tools** | PennyLane, PyTorch, torchvision |
| **Builds on** | Module 7 of the QML Lab | **Workspace** | `qml_projects/quantum-transfer-learning/` |

**Summary.** Transfer learning reuses a network trained on a large data set and retrains only a small final part for a new task. Here the reused part is a pretrained ResNet-18 and the retrained part is a "dressed" quantum circuit: a classical layer, a 4-qubit variational circuit and another classical layer. The project measures whether the quantum head matches classical heads of the same size, and how the result depends on qubit count and on the data set.

## 1 Background and key concepts

### 1.1 Transfer learning

Deep convolutional networks such as ResNet-18 (He et al., 2016) learn general visual features on ImageNet. Removing the final layer leaves a frozen feature extractor that maps an image to a vector $z \in \mathbb{R}^{512}$. Training only a new head on $z$ works well even with a few hundred labelled images.

### 1.2 Dressed quantum circuits

Mari et al. (2020) proposed classical-to-quantum transfer learning with a head of the form

$$\hat{y} = W_{\text{out}} \, \langle Z \rangle_\theta \Big( \tfrac{\pi}{2} \tanh \left( W_{\text{in}} z \right) \Big), \tag{1}$$

where $W_{\text{in}} \in \mathbb{R}^{n \times 512}$ reduces the features to $n$ angles, the variational circuit returns one $\langle Z \rangle$ per qubit, and $W_{\text{out}}$ maps them to class scores. Only the head is trained, with PyTorch (Paszke et al., 2019).

### 1.3 What a fair comparison looks like

The backbone does most of the work, so the comparison must hold everything else fixed: the same frozen features, the same training budget, and classical heads with a similar number of parameters (for example $512 \to n \to 2$ with a nonlinearity in place of the circuit).

## 2 Research questions

The project is open-ended: these questions are the starting point, not the finish line. Add new ones to the README and the journal as they come up.

**RQ1** Does the quantum head match a classical head with the same number of parameters?

**RQ2** How does accuracy change with 2, 4 and 6 qubits and with circuit depth?

**RQ3** How much does the $512 \to n$ reduction layer matter compared with the circuit itself?

## 3 Method

### 3.1 Data

The two-class ants/bees image set used by Mari et al. (2020) (a few hundred images), then a two-class subset of CIFAR-10. Standard ImageNet normalisation and light augmentation; fixed train/validation splits per seed.

### 3.2 Models

Frozen ResNet-18 from torchvision; dressed quantum circuit with $n = 4$ qubits, Hadamard layer, angle encoding and depth 6 of $R_Y$ rotations and CNOT chains; cross-entropy, Adam, 15 epochs, batch size 4.

### 3.3 Baselines

Linear head $512 \to 2$; MLP head $512 \to n \to 2$ with tanh (same parameter count as the dressed circuit apart from the circuit angles); fine-tuning the whole network as an upper bound.

### 3.4 Metrics

- Validation accuracy (mean $\pm$ s.d. over seeds).
- Number of trainable head parameters; training time per epoch.

## 4 Planned experiments

Each experiment becomes one `[[experiments]]` entry in the workspace, so its runs, parameters and metrics are stored and compared in the portal.

## 5 Running it in the QML Lab

The workspace lives in `qml_projects/quantum-transfer-learning/`. Scripts import three helpers from `qml_lab/plots.py`:

- `params()` reads the parameters from the portal's run form (defaults when run by hand);
- `save()` stores figures with the run;

| ID | Experiment | What varies | Main metrics |
|---|---|---|---|
| E1 | Quantum vs classical head | head type, $n = 4$ | accuracy |
| E2 | Qubit sweep | $n \in \{2, 4, 6\}$ | accuracy |
| E3 | Circuit depth | depth 1..8 | accuracy, time |
| E4 | Noise | noisy simulator at inference | accuracy drop |
| E5 | Second data set | CIFAR-10 pair | accuracy |

Table 1: Experiment plan. Results go into the journal and the results log (section 9).

- `log_metrics()` records numbers that appear in the run table.

```
qml_projects/quantum-transfer-learning/
  project.toml          metadata + experiments
  README.md             goal, open questions, ideas
  experiments/          runnable scripts
  docs/main.tex         this document
```

First experiment: This experiment is *planned*: write the script, add the block below to `project.toml`, then run `python manage.py sync_projects`.

```
[[experiments]]
slug = "ants-bees"
title = "Dressed quantum head on ResNet-18 features"
script = "experiments/e01_ants_bees.py"
figure = "head_comparison.png"

[experiments.params]
n_qubits = 4
q_depth = 6
epochs = 15
lr = 0.0004
batch_size = 4
seed = 0
```

```
# from the portal folder: pick up changes to project.toml
python manage.py sync_projects
cd qml_projects/quantum-transfer-learning
# run by hand with the default parameters
PYTHONPATH=../../qml_lab python experiments/e01_ants_bees.py
```

## 6 Quality bar

- A classical baseline with a similar number of trainable parameters, tuned with the same care (Bowles et al., 2024).
- At least five random seeds per configuration, reported as mean $\pm$ standard deviation.
- How results change with the number of qubits.
- At least one run under a realistic noise model.
- Compare heads only on identical frozen features; report head parameter counts.

## 7 Open directions

- Quantum-to-classical transfer: a pretrained quantum feature extractor with a classical head.
- Inference of the trained head on real hardware (Project 16).
- Other backbones (MobileNet, vision transformers).

## 8 Risks and limitations

- The backbone dominates; small differences between heads may be noise.
- Tiny data sets give high variance; use several seeds.

## 9 Results log

Summarise findings here as they become stable; day-to-day notes belong in the portal journal.

| Date | Run | Finding | Next step |
|---|---|---|---|

## References

Bowles, J., Ahmed, S. and Schuld, M. (2024) 'Better than classical? The subtle art of benchmarking quantum machine learning models', Preprint, arXiv:2403.07059. Available at: https://arxiv.org/abs/2403.07059.

He, K., Zhang, X., Ren, S. and Sun, J. (2016) 'Deep residual learning for image recognition', in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, pp. 770–778. Available at: https://doi.org/10.1109/CVPR.2016.90.

Mari, A., Bromley, T. R., Izaac, J., Schuld, M. and Killoran, N. (2020) 'Transfer learning in hybrid classical-quantum neural networks', *Quantum*, 4, 340. Available at: https://doi.org/10.22331/q-2020-10-09-340.

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J. et al. (2019) 'PyTorch: An imperative style, high-performance deep learning library', in *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, pp. 8024–8035.